Thao Nguyen Bui

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Education

DePauw University — Greencastle, IN

Bachelor of Arts in Computer Science and Mathematics | GPA: 3.81, Summa Cum Laude — *Aug. 2022 – May 2026*

- **Awards:** 2nd Place – Indiana Collegiate Mathematics Competition; WiCyS 2025 Conference Scholar; Dean's List
- **Coursework:** Object-Oriented Programming (Java), Data Structures & Algorithms (C++), Database and File Systems (SQL), Computer Systems (C), Mobile Development (Kotlin), Statistical Computing, Linear Algebra
- **Affiliate:** Grace Hopper Conference, SHPE, Tapia Conference, Rewriting The Code, WECode, Women In CyberSecurity
- **Leadership:** Vice-President at DePauw Vietnamese Student Association, Public Relations Chair at DePauw Maths Club

Experience

Pay Your Tuition Foundation — Washington, D.C.

Software Engineer Intern — *Aug. 2025 – Dec. 2025*

- Developed and maintained a student crowdfunding and tuition-support platform serving 2000+ students using **Webflow**, **Memberstack**, **Salesforce**, **Make**, and **REST APIs** to synchronize application, campaign, donor, and payment data.
- Built **12+** eligibility, fraud-review, approval, and reconciliation workflows using **JavaScript**, **Salesforce**, **Webflow**, **REST APIs** and **webhooks**, increasing throughput from 70% to 95% and reducing manual QA by 2+ hours daily.
- Migrated **Stripe to Razorpay** for tracking **$10M+** in tuition funds, improving webhook and payment reconciliation.
- Built and stabilized **REST API integrations** using **JSON, Salesforce, Make, and Vee AI**, resolving 50+ integration issues, cutting sync failures from 40% to 3%, and automating grant summaries, donor outreach, and follow-ups.

Wayfair (via Extern) — Remote

AI Agent Engineering Extern — *Jul. 2026 – Sep. 2026*

- Engineered a **4-agent pipeline** in **n8n** — Market Trend Discovery, Competitor Intelligence, Content Strategy, and Dashboard Builder — using **Mistral AI**, **Google Gemini**, **HuggingFace FLUX**, **Cheerio** and **Google Drive APIs**.
- Built a **Market Trend Discovery Agent** scanning Amazon listings, Instagram, Pinterest, and design blogs to detect emerging style signals and surface micro-segments **6–12 months ahead** of traditional sales data.
- Built a **Competitor Intelligence Agent** pulling live pricing, assortment, and feature data across competitors to reduce manual competitive analysis from hours to **20 minutes per execution**.

VietinBank — Da Nang, Vietnam

Software Engineer Intern — *May 2025 – Aug. 2025*

- Engineered transaction-validation and account-lookup services with **Java Spring Boot**, **Oracle SQL**, and **REST APIs**, enabling balance and transaction retrieval within 10 seconds and improving data access by 35%.
- Built a **Python** and **SQL** reconciliation pipeline to detect duplicate transactions, missing records, and balance mismatches across 50+ daily records, reducing accountants' manual review workload by 80%.
- Developed **reporting APIs** and reusable **data-processing modules** that delivered validated transaction and balance data to internal dashboards and monthly reports, reducing errors and accelerating issue investigation.

Projects

ProofVault - Blockchain Verification Platform | *React, TypeScript, FastAPI, PostgreSQL, Polygon, AWS, Docker*

- Engineered a **full-stack evidence verification system** that organizes files by case, generates **SHA-256 fingerprints**, records immutable proofs on **Polygon**, and detects whether evidence has been modified.
- Developed 11 **FastAPI endpoints**, including **presigned-upload and file-verification APIs**, with ownership validation, transactional database operations, **SQLAlchemy models**, **Alembic migrations**, and 14 automated backend and smart-contract tests.
- Designed a **Dockerized AWS architecture** with private **S3 storage**, least-privilege **IAM**, **ECR image management**, **ECS/Fargate orchestration**, **ALB health checks**, **CloudWatch logging**, and **Amplify** deployment.

KITES - Android Family-safety App | *React Native, TypeScript, Node.js, Express.js, REST APIs, Recoil*

- Built and deployed an **Android family-safety app**, leveraging **React Native** for mobile functionality, **Express.js** for backend services, and **React Native Maps** for location tracking, guardian management, and emergency communication.
- Designed a modular architecture with **TypeScript**, **REST APIs**, **Recoil**, **AsyncStorage**, and centralized service layers, supporting 10+ parent-facing safety features including location tracking, guardian management, emergency alerts, and **Gemini-powered chatbot** guidance while simplifying maintenance and debugging with **Claude Code**.

Technical Skills

Programming Languages: Java, Python, C++, JavaScript, TypeScript, SQL, R, HTML/CSS
Frameworks & Libraries: React, React Native, Node.js, Express.js, Spring Boot, Axios, AWT, Firebase
Databases: MongoDB, PostgreSQL, Oracle SQL, MySQL, SQLite
Tools & Services: Git, AWS, GCP, Azure, WebSocket, Salesforce, Webflow, Razorpay, Linux/Unix, TensorFlow